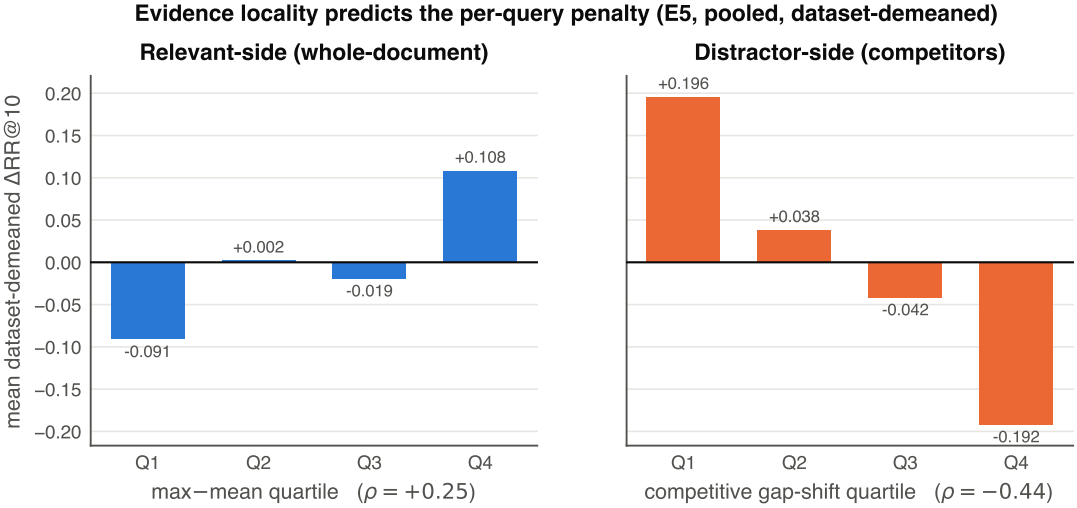
Evidence locality predicts the per-query penalty (E5, pooled, dataset-demeaned)
Relevant-side (whole-document)
Distractor-side (competitors)
mean dataset-demeaned ΔRR@10
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
+0.108
+0.002
-0.019
-0.091
+0.196
+0.038
-0.042
-0.192
Q1
Q2
Q3
Q4
max−mean quartile (ρ = +0.25)
competitive gap-shift quartile (ρ = −0.44)